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Roll No:-2401038
AI and DS
Division:T1
Batch:T12
Subject: ASD & DevOps

EXPERIMENT:-7

Aim: To implement monitoring using Prometheus / Grafana / Nagios.

LO mapped: **LO5**

LO5: Implement ML lifecycle management and automated workflows using MLflow, Airflow, and monitoring tools like Prometheus/Grafana/Nagios.

Theory:

Continuous monitoring is the process of continuously observing servers, applications, networks, and other infrastructure components to identify performance issues, failures, and security-related problems. It plays an important role after an application or service is deployed because it helps administrators detect problems early and take corrective action.

Nagios is an open-source monitoring system used to monitor hosts, networks, applications, and services. It can detect issues such as host failures, high resource utilization, and service unavailability. Nagios also provides alerts and notifications, helping administrators troubleshoot problems and maintain system availability.

The major benefits of Nagios include:

- Detecting network and server failures.
- Monitoring the availability and performance of services.
- Identifying problems at an early stage.
- Helping determine the root cause of failures.
- Supporting infrastructure planning and upgrades.
- Sending alerts when monitored resources become unavailable.
- Improving the reliability and availability of IT infrastructure.

For monitoring Windows systems, NSClient++ can be installed on the Windows host. It acts as a remote monitoring agent and allows Nagios to execute checks such as CPU, memory, disk,

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service, and connectivity checks. NRPE (Nagios Remote Plugin Executor) is commonly used for executing monitoring plugins on remote systems.

Since Nagios Core is primarily designed for Linux, it can be deployed on Windows using Docker Desktop with WSL2. The Nagios Core container provides the monitoring server and web interface, while NSClient++ runs as a service on the Windows host being monitored.

Other monitoring tools include Prometheus, which collects and stores time-series metrics, and Grafana, which provides dashboards for visualizing those metrics. Together, these tools enable administrators to observe system behavior, analyze performance trends, and identify abnormal conditions.

In this experiment, Nagios Core is run inside a Docker container, the Nagios web interface is configured, NSClient++ is installed on the Windows host, and the Windows host is added to Nagios. A PING service is configured to continuously check the availability of the host.

Output:

```
Last login: Tue Sep  8 10:17:31 on ttys000
kyra@Kyras-MacBook-Air ~ % docker --version
Docker version 29.7.2, build a7dcaa6
kyra@Kyras-MacBook-Air ~ % docker version
Client:
 Version:           29.7.2
 API version:       1.55
 Go version:        go1.26.5
 Git commit:        a7dcaa6
 Built:             Wed Aug  5 18:27:50 2026
 OS/Arch:           darwin/arm64
 Context:           desktop-linux

Server: Docker Desktop 4.90.0 (238679)
 Engine:
  Version:          29.7.2
  API version:      1.55 (minimum version 1.40)
  Go version:       go1.26.5
  Git commit:       6a43e3d
  Built:            Wed Aug  5 18:28:35 2026
  OS/Arch:          linux/arm64
  Experimental:     false
 containerd:
  Version:          v2.3.3
  GitCommit:        aad11006b869517fcd3009450b6f82da282e1a9b
 runc:
  Version:          1.4.3
  GitCommit:        v1.4.3-0-gbb14dabe
 docker-init:
  Version:          0.19.0
```

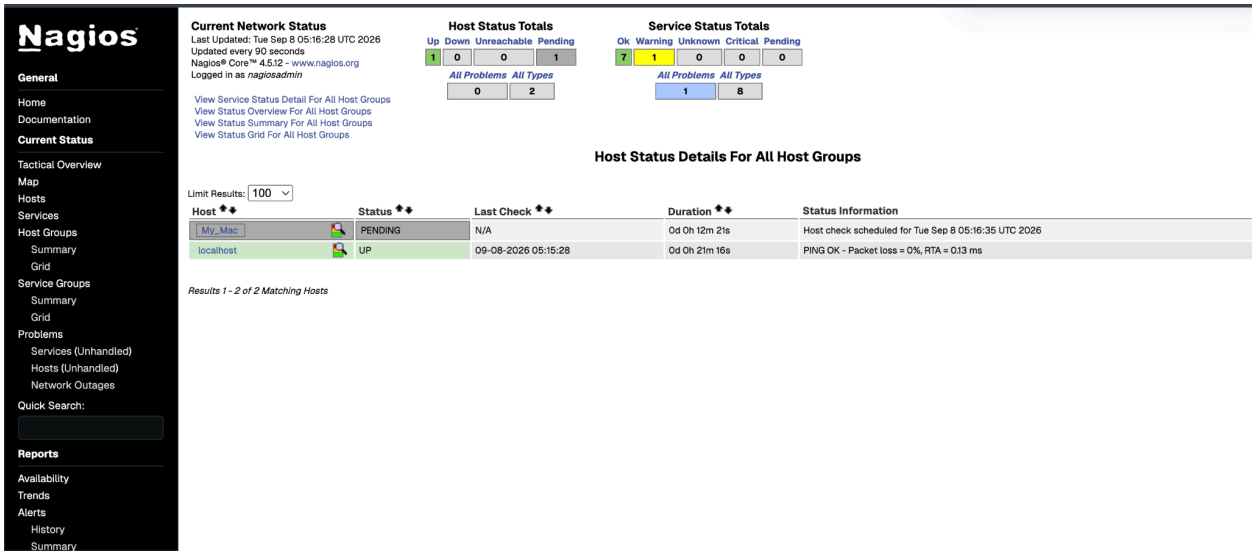
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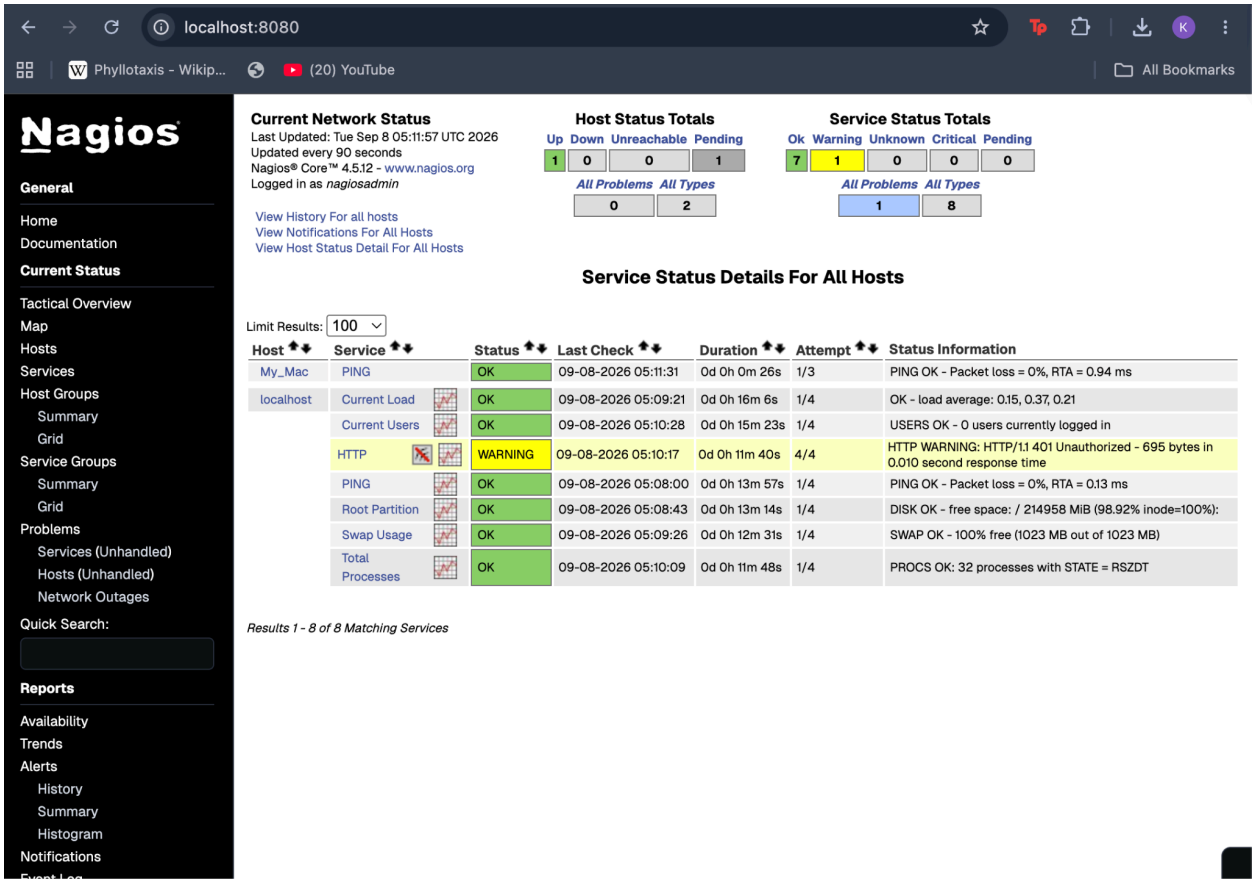
```
kyra@Kyras-MacBook-Air ~ % docker pull jasonrivers/nagios:latest
latest: Pulling from jasonrivers/nagios
cb259a83ac3d: Pull complete
7a7447982198: Pull complete
59215b8e47cf: Pull complete
6fe99ef1bed3: Pull complete
fdbaa15b8986: Pull complete
50ea270d34e9: Pull complete
80b6ab7aef5d: Pull complete
254354b70ac0: Pull complete
f47c3af6bf44: Pull complete
a03720c2c16b: Pull complete
9288c4992679: Pull complete
748b632e75e7: Pull complete
54b9e993e315: Pull complete
0d59a81a5608: Pull complete
1b9771f55a12: Pull complete
2908452ff193: Pull complete
a7bd6dce6a77: Pull complete
1413dee23aae: Pull complete
dc3a88012a3e: Pull complete
7edde1931fd2: Pull complete
a80db686d641: Pull complete
37f75c8146e2: Pull complete
e68fcda12c0c: Pull complete
c5701b5e1540: Pull complete
d003d4303177: Pull complete
f2f9a1b3ce48: Pull complete
c42073aaf047: Pull complete
a6b90b5bbb15: Pull complete
e4dc1d4e010c: Pull complete
Digest: sha256:fee9cd6a2933e790dde613d70c3a603f96978d65fa6bbc3d617fce807e00f2e2
Status: Downloaded newer image for jasonrivers/nagios:latest
docker.io/jasonrivers/nagios:latest
```

```
kyra@Kyras-MacBook-Air ~ % docker ps
CONTAINER ID   IMAGE                     COMMAND                  CREATED         STATUS         PORTS                                       NAMES
afd0c0cb7ca4   jasonrivers/nagios:latest   "/usr/local/bin/star…"   13 seconds ago   Up 13 seconds   0.0.0.0:8080->80/tcp, [::]:8080->80/tcp   nagios4
```

```
kyra@Kyras-MacBook-Air ~ % docker exec -it nagios4 htpasswd -c /opt/nagios/etc/htpasswd.users nagiosadmin
New password:
Re-type new password:
Adding password for user nagiosadmin
```

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Conclusion: The experiment successfully demonstrated the implementation of continuous infrastructure monitoring using Nagios. Nagios Core was deployed through Docker, a Windows host was configured with NSClient++, and the host was added to Nagios for monitoring. The PING service was used to verify host availability through the Nagios web interface. This experiment helped in understanding how monitoring tools can detect infrastructure problems and improve system reliability, availability, and performance.

LO5 mapped successfully.